

V3 moneyness performance

Same 6 models · 3 companies · 12 windows · 7-day hourly study · classified by call moneyness $m = S/K$

What this report is

- Companion to V3_7d_hourly_empirical_study. The notebook in this folder is the source of truth; this PDF is written from the same moneyness payload.
- Question: does model ranking change with option moneyness, overall and inside each volatility regime?
- BEST in every table is the lowest RMSE%. That row is bold and green, and the Mark column says BEST. Empty buckets ($n = 0$) show em dashes and have no BEST.

Unchanged experimental design

- Same six models, three companies (SPY, AAPL, MSFT), 12 windows, 7-day lookback, hourly recalibration, V3 filters, Euler-corrected Heston step, LSM $n_{\text{paths}} = 10000$ / seed 42.
- Same option contracts as the main study (nearest-ATM listed calls, DTE 7-60, $|S/K - 1| \leq 10\%$). No model-specific re-sampling.
- Pooled $n = 1224$ contracts per model (identical keys across models).

Moneyness rule (calls, $m = S / K$)

- DOTM: $m < 0.98$. OTM: $0.98 \leq m < 0.995$. ATM: $0.995 \leq m \leq 1.005$. ITM: $1.005 < m \leq 1.02$. DITM: $m > 1.02$.
- These five bins partition the main-study filter $|m - 1| \leq 10\%$. The Monday nearest-ATM design puts most mass in ATM; DOTM and DITM are sparse and appear mainly in 2008-2009.

Metrics

- $\text{RMSE\%} = 100 \times \sqrt{\text{mean}(((C_{\text{model}} - C_{\text{mkt}})/C_{\text{mkt}})^2)}$. Ranking uses this only.
- MAE and Bias \$ in option-premium dollars. $\text{Bias\%} = 100 \times \text{mean}((C_{\text{model}} - C_{\text{mkt}})/C_{\text{mkt}})$. Early-exercise fraction = mean LSM share of paths that stop before expiry.

Two analyses

- Analysis 1 — Overall moneyness: pool all companies and regimes. Five tables (one per bucket) plus a Model $\times$ Moneyness RMSE% heatmap.
- Analysis 2 — Window $\times$ moneyness: keep the 12 windows separate, pool the three companies. Compact RMSE tables/heatmaps and a winner grid showing whether the best model changes with moneyness inside each window.

Table M0 · Contract counts (one model = all models)

Regime	DOTM	OTM	ATM	ITM	DITM	Total
2022-10-21 Oct 2022	0	9	19	28	46	102
2022-11-18 Nov 2022	53	16	6	6	21	102
2022-12-16 Dec 2022	37	18	19	14	14	102
2023-01-20 Jan 2023	66	2	0	2	32	102
2023-02-17 Feb 2023	59	20	7	2	14	102
2023-03-17 Mar 2023	2	53	24	23	0	102
2023-04-21 Apr 2023	38	32	19	10	3	102
2023-05-19 May 2023	90	12	0	0	0	102
2023-06-16 Jun 2023	102	0	0	0	0	102
2023-07-21 Jul 2023	102	0	0	0	0	102
2023-08-18 Aug 2023	102	0	0	0	0	102
2023-09-15 Sep 2023	102	0	0	0	0	102
All regimes	753	162	94	85	130	1224

Moneyiness rule · American calls, $m = S/K$

Bucket	Rule	n (pooled)	Share
DOTM	$m < 0.98$	753	61.5%
OTM	$0.98 \leq m < 0.995$	162	13.2%
ATM	$0.995 \leq m \leq 1.005$	94	7.7%
ITM	$1.005 < m \leq 1.02$	85	6.9%
DITM	$m > 1.02$	130	10.6%

ATM dominates because the main study picks the nearest-ATM listed call each Monday. DOTM/DITM are not empty because of a new filter — they are rare in that ATM sample, especially after 2009. Every model still sees the same contracts in every bucket.

Page 3 · Analysis 1 · Overall moneyness · Tables M1-M3

Pooled SPY + AAPL + MSFT and all 12 windows. Bold green = lowest RMSE%.

Table M1 · DOTM · $m < 0.98$ · $n = 753$ · BEST = Heston

Model	n	RMSE%	MAE	Bias \$	Bias%	Early-ex.	Mark
GBM	753	99.95	7.0573	-7.0573	-99.95	0.2%	
GARCH	753	99.92	7.0552	-7.0552	-99.92	0.4%	
Heston	753	98.34	6.9416	-6.9416	-98.25	2.3%	BEST
Merton	753	99.95	7.0572	-7.0572	-99.95	0.2%	
GARCH-Merton	753	99.91	7.0544	-7.0544	-99.90	0.4%	
Heston-Merton	753	98.34	6.9417	-6.9417	-98.25	2.2%	

Table M2 · OTM · $0.98 \leq m < 0.995$ · $n = 162$ · BEST = Heston-Merton

Model	n	RMSE%	MAE	Bias \$	Bias%	Early-ex.	Mark
GBM	162	97.86	6.8326	-6.8326	-97.83	6.4%	
GARCH	162	97.80	6.8243	-6.8243	-97.76	7.0%	
Heston	162	80.34	5.5982	-5.5982	-79.20	20.1%	
Merton	162	97.84	6.8317	-6.8317	-97.81	6.5%	
GARCH-Merton	162	97.45	6.8010	-6.8010	-97.40	7.5%	
Heston-Merton	162	80.22	5.5911	-5.5911	-79.08	19.9%	BEST

Table M3 · ATM · $0.995 \leq m \leq 1.005$ · $n = 94$ · BEST = Heston-Merton

Model	n	RMSE%	MAE	Bias \$	Bias%	Early-ex.	Mark
GBM	94	86.98	6.3900	-6.3900	-86.62	34.2%	
GARCH	94	86.87	6.3710	-6.3710	-86.53	34.2%	
Heston	94	61.18	4.3230	-4.3230	-58.56	36.5%	
Merton	94	86.96	6.3887	-6.3887	-86.60	33.1%	
GARCH-Merton	94	86.15	6.3187	-6.3187	-85.79	32.7%	
Heston-Merton	94	61.06	4.3110	-4.3110	-58.45	35.9%	BEST

Bias \$ = mean(model – market). Bias% = 100 × mean((model – market)/market).

Page 4 · Analysis 1 · Overall moneyness · Tables M4-M5

Pooled SPY + AAPL + MSFT and all 12 windows. Bold green = lowest RMSE%.

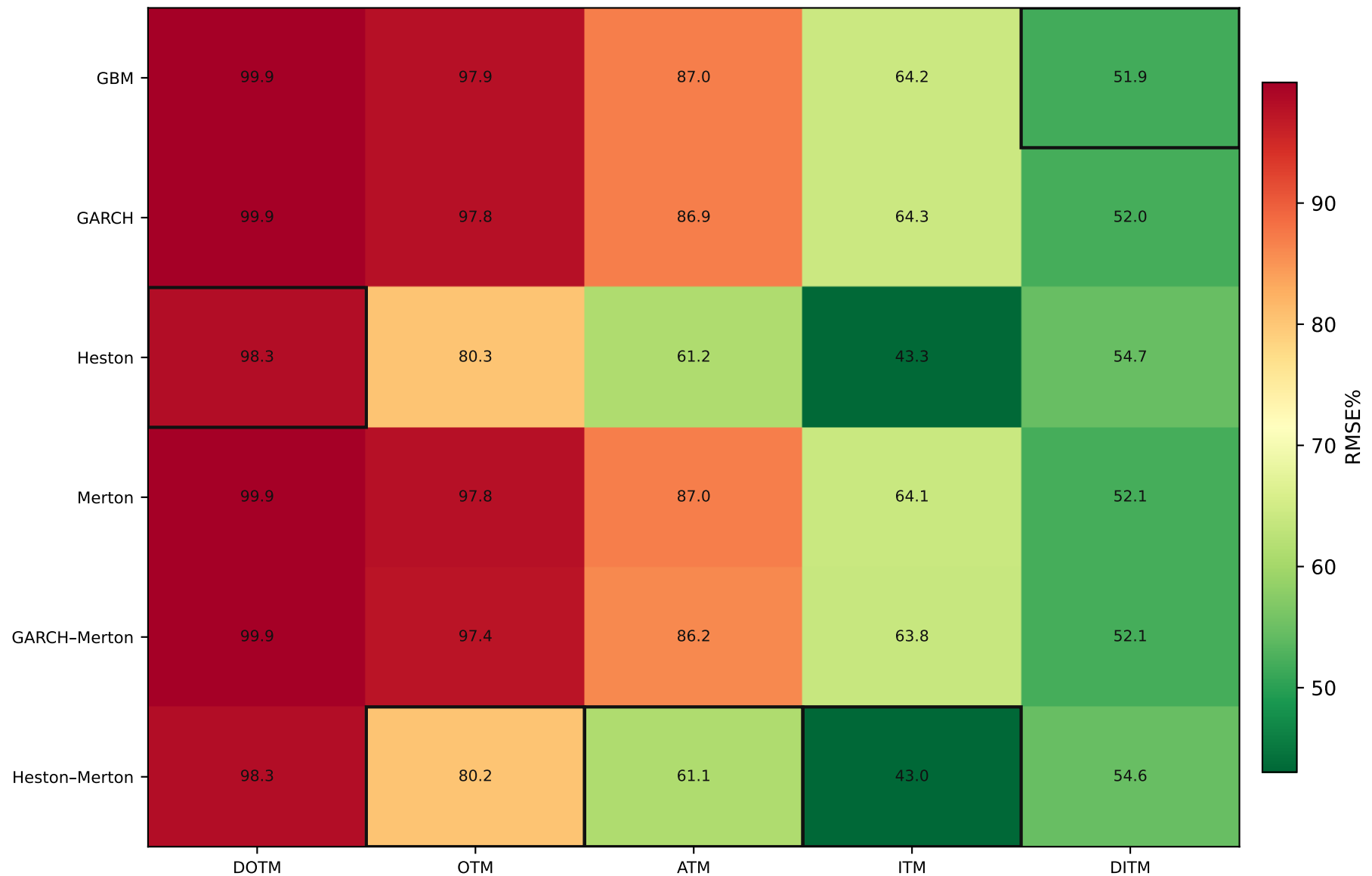
Table M4 · ITM · $1.005 < m \leq 1.02$ · $n = 85$ · BEST = Heston-Merton

Model	n	RMSE%	MAE	Bias \$	Bias%	Early-ex.	Mark
GBM	85	64.22	4.7075	-4.7075	-62.84	72.8%	
GARCH	85	64.29	4.7078	-4.7078	-62.94	73.2%	
Heston	85	43.27	3.0360	-3.0215	-39.60	51.9%	
Merton	85	64.14	4.7003	-4.7003	-62.75	68.3%	
GARCH-Merton	85	63.85	4.6733	-4.6733	-62.45	67.7%	
Heston-Merton	85	43.03	3.0186	-3.0073	-39.43	50.8%	BEST

Table M5 · DITM · $m > 1.02$ · $n = 130$ · BEST = GBM

Model	n	RMSE%	MAE	Bias \$	Bias%	Early-ex.	Mark
GBM	130	51.91	3.2890	+0.0883	+6.06	91.2%	BEST
GARCH	130	52.02	3.2966	+0.0940	+6.15	90.7%	
Heston	130	54.66	3.2349	+0.8055	+15.59	74.8%	
Merton	130	52.05	3.3004	+0.1033	+6.28	88.1%	
GARCH-Merton	130	52.13	3.2993	+0.1072	+6.35	88.9%	
Heston-Merton	130	54.63	3.2260	+0.8142	+15.69	73.4%	

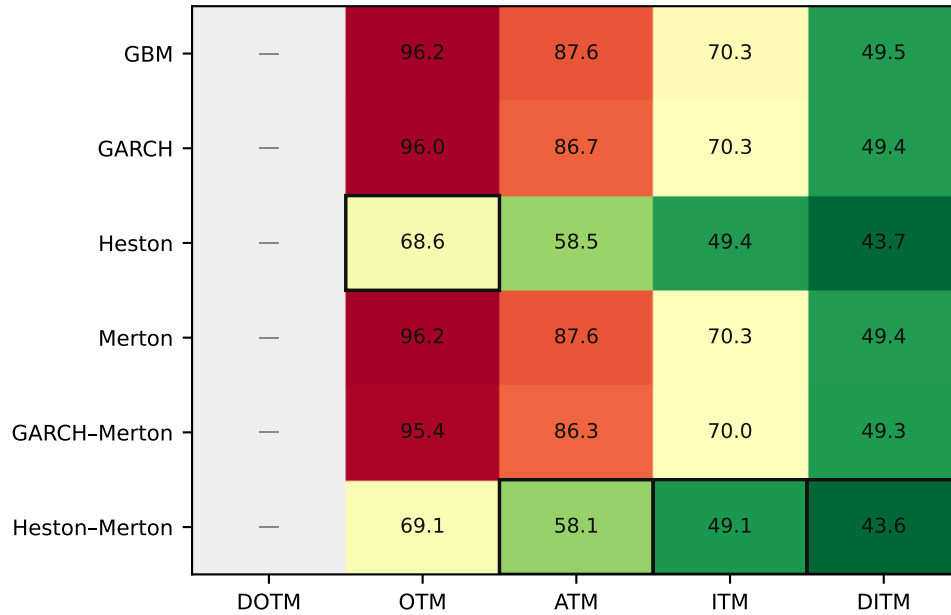
ITM/DITM: $S > K$. DOTM/OTM: $S < K$. ATM: $|S/K - 1| \leq 0.5\%$.



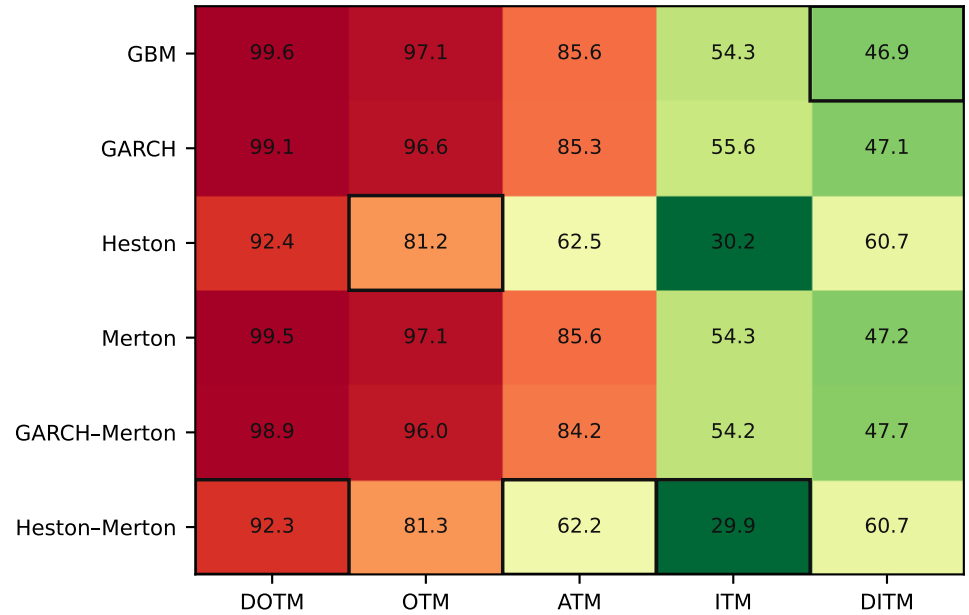
Black outline = lowest RMSE% in that moneyness column. Grey = n = 0. Pooled across companies and regimes.

Figure M2 · RMSE% by model × moneyness · each window (companies pooled)

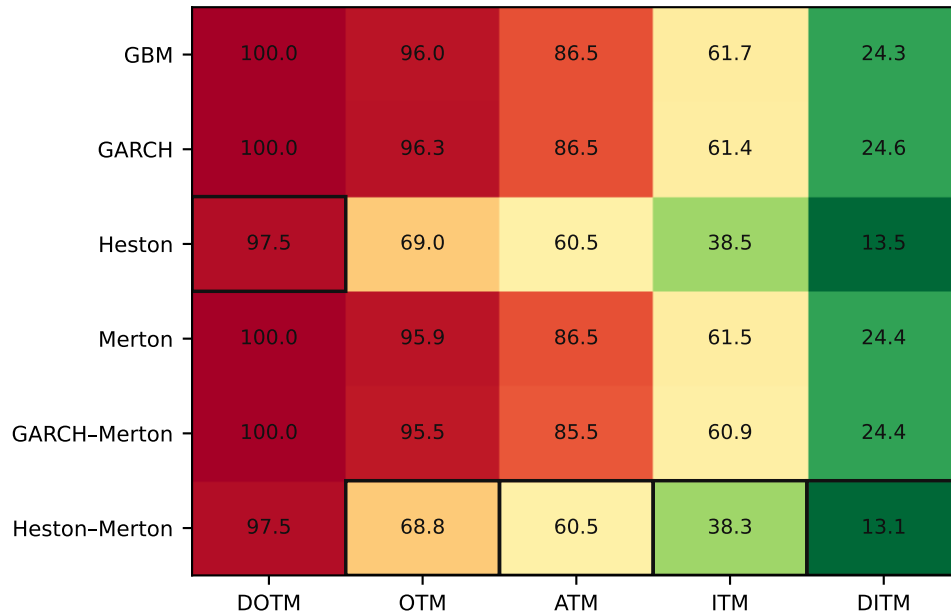
2022-10-21 · Oct 2022



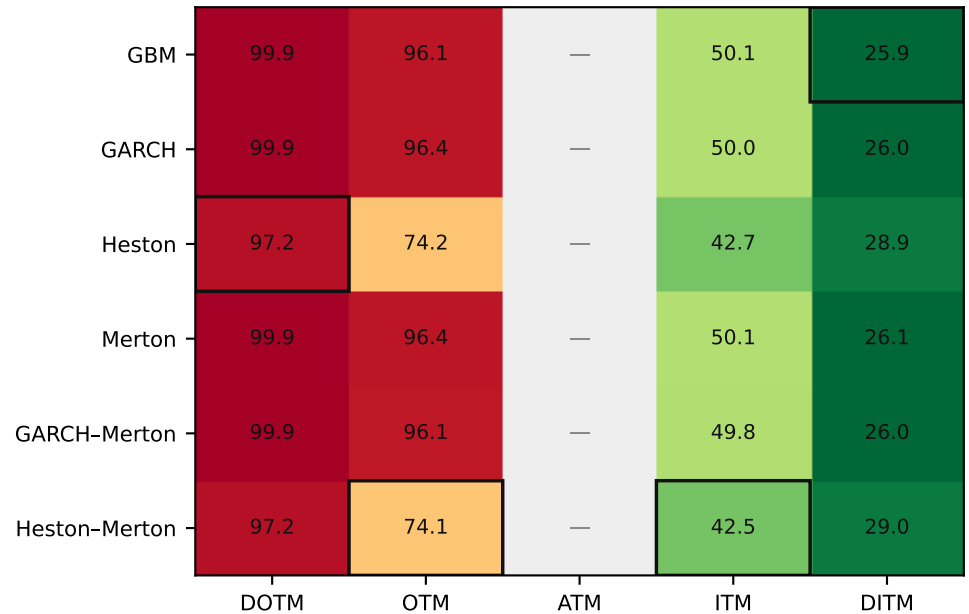
2022-11-18 · Nov 2022



2022-12-16 · Dec 2022



2023-01-20 · Jan 2023



Outlined cell = lowest RMSE% in that window × moneyness column. Grey = n = 0.

Figure M2 · RMSE% by model × moneyness · each window (companies pooled)

2023-02-17 · Feb 2023

GBM	99.9	97.4	75.1	41.2	106.2
GARCH	99.9	97.9	78.0	43.5	106.5
Heston	95.9	81.7	43.7	16.3	116.6
Merton	99.9	97.3	75.0	41.0	106.6
GARCH-Merton	99.9	97.5	76.5	41.3	106.7
Heston-Merton	95.9	81.6	44.2	15.1	116.6
	DOTM	OTM	ATM	ITM	DITM

2023-03-17 · Mar 2023

GBM	100.0	98.4	85.6	59.6	—
GARCH	100.0	98.0	85.0	59.8	—
Heston	93.3	78.1	56.1	33.1	—
Merton	100.0	98.4	85.6	59.5	—
GARCH-Merton	100.0	97.8	84.4	59.3	—
Heston-Merton	93.2	77.8	56.1	32.8	—
	DOTM	OTM	ATM	ITM	DITM

2023-04-21 · Apr 2023

GBM	100.0	99.2	92.9	71.3	36.5
GARCH	100.0	99.2	93.1	71.3	36.5
Heston	97.2	88.9	74.3	59.4	35.0
Merton	100.0	99.2	92.8	71.2	36.5
GARCH-Merton	100.0	99.1	92.6	71.2	36.6
Heston-Merton	97.2	88.8	74.1	59.2	34.9
	DOTM	OTM	ATM	ITM	DITM

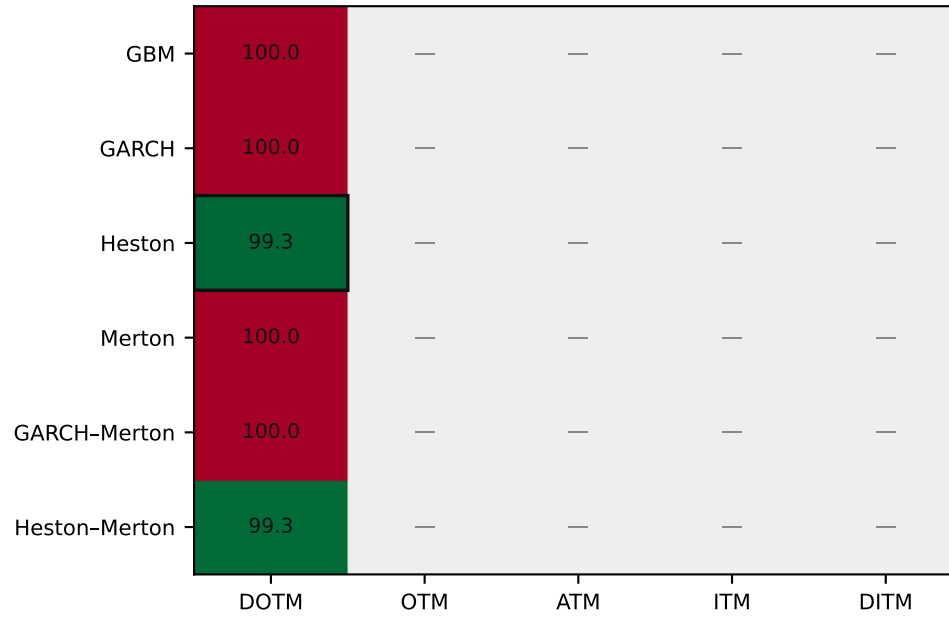
2023-05-19 · May 2023

GBM	100.0	98.0	—	—	—
GARCH	100.0	98.4	—	—	—
Heston	99.3	87.4	—	—	—
Merton	100.0	97.9	—	—	—
GARCH-Merton	100.0	98.1	—	—	—
Heston-Merton	99.2	87.3	—	—	—
	DOTM	OTM	ATM	ITM	DITM

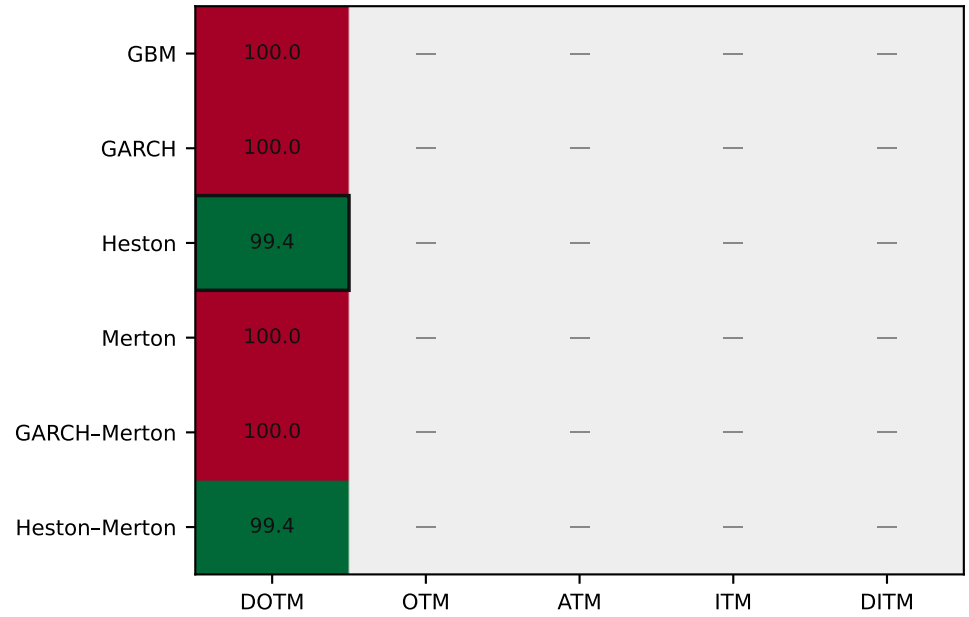
Outlined cell = lowest RMSE% in that window × moneyness column. Grey = n = 0.

Figure M2 · RMSE% by model × moneyness · each window (companies pooled)

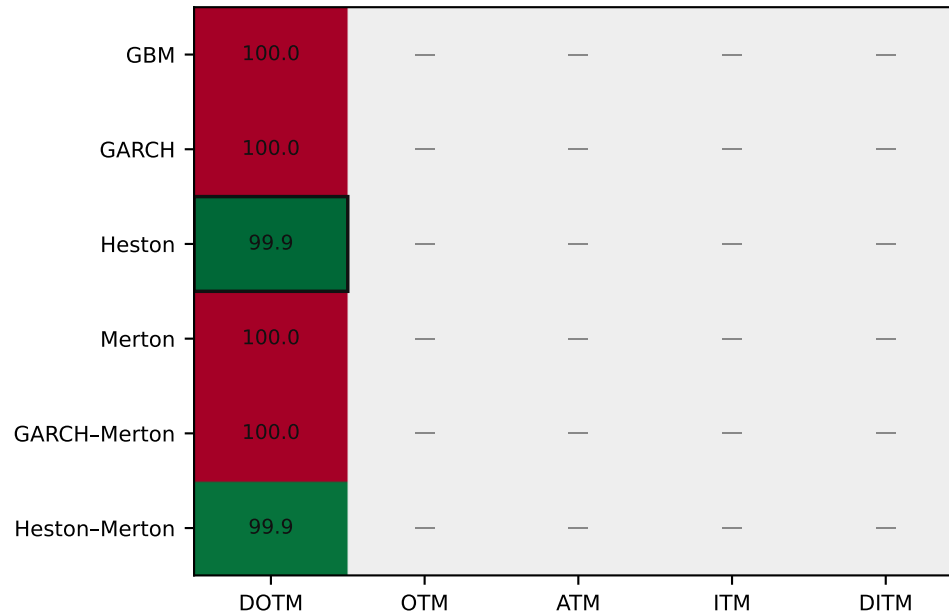
2023-06-16 · Jun 2023



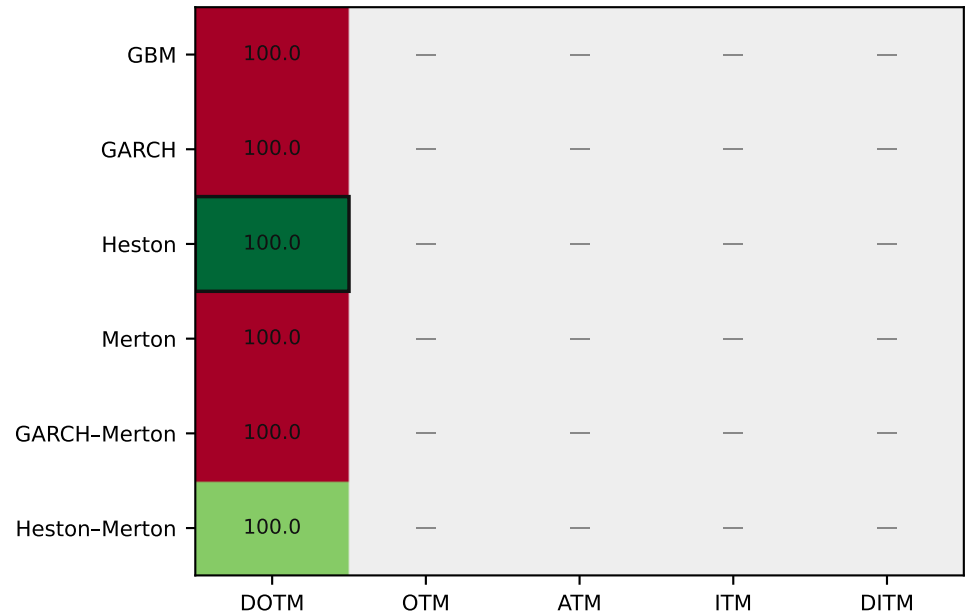
2023-07-21 · Jul 2023



2023-08-18 · Aug 2023



2023-09-15 · Sep 2023



Outlined cell = lowest RMSE% in that window × moneyness column. Grey = n = 0.

Analysis 2 · Does the ranking change with moneyness inside each window?

Table M6 · Best model (lowest RMSE%) by window × moneyness

Window	DOTM	OTM	ATM	ITM	DITM
2022-10-21 Oct 2022	—	Heston	Heston–Merton	Heston–Merton	Heston–Merton
2022-11-18 Nov 2022	Heston–Merton	Heston	Heston–Merton	Heston–Merton	GBM
2022-12-16 Dec 2022	Heston	Heston–Merton	Heston–Merton	Heston–Merton	Heston–Merton
2023-01-20 Jan 2023	Heston	Heston–Merton	—	Heston–Merton	GBM
2023-02-17 Feb 2023	Heston	Heston–Merton	Heston	Heston–Merton	GBM
2023-03-17 Mar 2023	Heston–Merton	Heston–Merton	Heston	Heston–Merton	—
2023-04-21 Apr 2023	Heston	Heston–Merton	Heston–Merton	Heston–Merton	Heston–Merton
2023-05-19 May 2023	Heston–Merton	Heston–Merton	—	—	—
2023-06-16 Jun 2023	Heston	—	—	—	—
2023-07-21 Jul 2023	Heston	—	—	—	—
2023-08-18 Aug 2023	Heston	—	—	—	—
2023-09-15 Sep 2023	Heston	—	—	—	—

Table M7 · n contracts by window × moneyness (shared across models)

Window	DOTM	OTM	ATM	ITM	DITM
2022-10-21 Oct 2022	0	9	19	28	46
2022-11-18 Nov 2022	53	16	6	6	21
2022-12-16 Dec 2022	37	18	19	14	14
2023-01-20 Jan 2023	66	2	0	2	32
2023-02-17 Feb 2023	59	20	7	2	14
2023-03-17 Mar 2023	2	53	24	23	0
2023-04-21 Apr 2023	38	32	19	10	3
2023-05-19 May 2023	90	12	0	0	0
2023-06-16 Jun 2023	102	0	0	0	0
2023-07-21 Jul 2023	102	0	0	0	0
2023-08-18 Aug 2023	102	0	0	0	0
2023-09-15 Sep 2023	102	0	0	0	0

A change in the Table M6 entry across a row means the ranking depends on moneyness inside that window. Columns with n = 0 cannot change the ranking.

Does performance change with moneyness?

- DOTM (n = 753): BEST = Heston with RMSE% = 98.34, Bias% = -98.25.
- OTM (n = 162): BEST = Heston-Merton with RMSE% = 80.22, Bias% = -79.08.
- ATM (n = 94): BEST = Heston-Merton with RMSE% = 61.06, Bias% = -58.45.
- ITM (n = 85): BEST = Heston-Merton with RMSE% = 43.03, Bias% = -39.43.
- DITM (n = 130): BEST = GBM with RMSE% = 51.91, Bias% = +6.06.
- Overall, the ranking does change with moneyness: DOTM → Heston, OTM → Heston-Merton, ATM → Heston-Merton, ITM → Heston-Merton, DITM → GBM.
- Inside regimes, the BEST model also changes with moneyness in: 2022-10-21, 2022-11-18, 2022-12-16, 2023-01-20, 2023-02-17, 2023-03-17, 2023-04-21.
- These comparisons use the hourly nearest-ATM sample (09:59-15:59; LSM skipped at Friday 15:59). ATM still holds most of the mass; sparse DOTM/DITM buckets should be read as descriptive.